

PROJECT PROPOSAL: NEXUS

A Multimodal Communication & Healthcare Intelligence Platform

GROUP 109 MEMBERS

NAME	INDEX NUMBER	REFERENCE
NANA KWADWO ADU ADJEI	6129124	21143082
ACHEAMPONG DANIEL NTI	6125424	21142799
Papa Kobina Asmah Okota	6174424	21142789
Emmanuel Narh	6170124	21142823
ALBERT OSEI	6176824	21142819

0. APP OVERVIEW — Nexus

Nexus is an integrated digital platform designed to solve communication gaps within the deaf and hard-of-hearing community, specifically focused on **Healthcare and Emergency Response**. By utilizing real-time multimodal translation, Nexus converts signs into speech and transforms typed words or spoken audio into high-fidelity sign animations.

This comprehensive solution features a **Health Ticker** for real-time medical news, embedded **First-Aid Video Tutorials** with integrated sign language, and a robust **Linguistic Translation Engine** that ensures all parties understand the information in their preferred spoken or written language before conversion.

1. PROBLEM STATEMENT

Information barriers in healthcare lead to severe medical complications for the deaf community. Often, medical news and emergency first-aid videos are not accessible because they lack visual sign-language embedding or are provided in a language the user does not understand.

Key Challenges

- **The Translation Gap:** Standard tools lack the ability to translate between spoken languages before converting to sign language.
- **Medical Vulnerability:** Lack of real-time health news and accessible emergency first-aid resources for deaf users.
- **Inconsistent Access:** Information is scattered across platforms rather than unified in a single, safe "Nexus."

2. LINGUISTIC ALIGNMENT & CROSS-TRANSLATION

A unique feature of Nexus is the primary linguistic alignment layer. Before any multimodal conversion occurs, the system ensures the underlying text or speech is translated into a language understood by the user (e.g., translating medical news from English to a local language or another international language like French).

- **Contextual Understanding:** Uses NLP to ensure medical terminology is translated accurately across spoken languages.
- **Dual-Layer Translation:** *Input (Language A) → Translation (Language B) → Final Output (Sign Animation/Speech)*.

3. APPROACH & SOLUTION

Nexus utilizes a **Microservice-driven approach** to provide a seamless front-end experience for critical tasks.

- **Healthcare News Portal:** Real-time updates appearing on the dashboard via an integrated news service.
- **First-Aid Library:** Embedded videos for emergency procedures featuring integrated sign-language overlays.
- **Speech & Text to Sign:** Converts external spoken medical advice or typed text into 3D animated signs.
- **Offline Capability:** Critical first-aid resources cached locally on the device for internet-free access.

4. SYSTEM ACTORS

Actor	Role
Primary User	Deaf or hard-of-hearing individual accessing translation and health resources.
Medical Personnel	Provides spoken or typed medical instructions to be translated for the user.
News/Admin	Updates the healthcare ticker and uploads new first-aid video content.

5. DESIGN & SYSTEM ARCHITECTURE

The system is built on a **Microservice Architecture** to ensure that the Health News service, Translation service, and Media service remain independent and scalable.

Infrastructure Tools

- **Frontend:** Expo (TypeScript) — for a responsive, cross-platform mobile app.
- **Backend:** Spring Boot (Java) — powering the individual microservices.
- **Database:** PostgreSQL — for relational data storage and historical logs.
- **Gateway:** API Gateway for routing requests between the mobile app and microservices.

6. REQUIREMENTS

6.1 Functional Requirements

- **Multimodal Engine:** Sign gesture recognition and speech/text-to-sign conversion.
- **Health Ticker:** Front-end display of live medical news and alerts.
- **Video Module:** Playback of embedded first-aid tutorials with sign language embedding.
- **Cross-Language Support:** Translation between written/spoken languages.

6.2 Non-Functional Requirements

- **Type Safety:** Enforced through TypeScript on the mobile client.
- **Scalability:** Enabled by the Microservice architecture.
- **Reliability:** Redundant PostgreSQL instances for data integrity.

7. BACKEND & DATABASE DESIGN

7.1 Microservice Responsibilities

Service	Responsibility
News Service	External API integration for live medical headlines.
Media Service	Serves embedded video content and manages file storage.
Translate Service	NLP logic for language-to-language and text-to-sign mapping.

7.2 Core Entities (PostgreSQL)

Entity	Attributes
Users	ID, Index_Number, Name, Medical_Preferences.
Health_Ticker	Entry_ID, Headline, Source, Date_Posted.
Video_Vault	Video_ID, Title, Category (First-Aid), Storage_Path.

8. API DESIGN & ENDPOINTS

Method	Endpoint	Description
POST	/auth/login	User identity verification.
GET	/health/news	Fetches current ticker headlines.
POST	/translate/multimodal	Converts speech/text to sign language data.
GET	/media/first-aid	Retrieves the list of available instructional videos.

9. TECHNOLOGY STACK

Nexus is built using a professional enterprise stack designed for scalability, security, and type-safety across all services.

- **Frontend (Mobile):** Expo TypeScript React Native
- **Backend (Logic):** Spring Boot (Java) Microservices Architecture
- **Database:** PostgreSQL
- **Architecture:** RESTful API API Gateway

10. FINAL SUMMARY

Nexus provides a comprehensive solution for the deaf community by unifying healthcare news, emergency first-aid, and multimodal translation into one resilient platform. By leveraging **Expo**, **Spring Boot**, and **PostgreSQL** within a microservice architecture, Nexus ensures that communication barriers are broken and vital health information is always accessible, regardless of language or medium.